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Lightweight RAW Object Detection for Automated Driving

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***Abstract*—Modern sensor and image acquisition technologies have enabled Automated Vehicles (AVs) to leverage image data for enhanced on-board visual perception. To achieve this, AVs deploy Computer Vision (CV) deep learning tools for safe and accurate object detection. These systems typically rely on Image Signal Processing (ISP) pipelines to convert RAW sensor data into human-perceivable RGB images, as state-of-the-art deep learning models are optimised for such inputs. However, in safety-critical real-time applications such as AVs, CV systems must not only be reliable and accurate but also fast. In this work, we demonstrate that a lightweight ISP comprising simple processing steps can effectively operate on RAW image data while maintaining strong object detection performance across diverse scenes. Our experiments show that our proposed simple ISP maintains a balanced trade-off between accuracy and latency.**

***Index Terms*—RAW, ISP, Automated Vehicles, Lightweight Processing, Signal Processing, Computer Vision**

I. Introduction

Automated Vehicles (AVs) aim to usher in a new era in the field of transportation by preventing and counteracting human errors which lead to traffic accidents and millions of road casualties every year [1]. AVs utilise various sensors – visual, radio, ultrasonic, and infrared – to capture different regions of the electromagnetic spectrum, and reliably perceive their surroundings for decision-making. The deployment of diverse sensors adds essential redundancy and improves system reliability by using the strengths of one sensor to compensate for the weaknesses of another [2]. AVs employ multiple automotive cameras on the front, rear and sides of the vehicle to capture visual data, which are either displayed on in-cabin monitors to augment driver awareness or are forwarded towards Computer Vision (CV) modules to support the Advanced Driver Assistance Systems (ADAS). A single camera produces a data stream which ranges between 500 Mb/s to 3.5 Gb/s [3].

The data captured by the image sensors are in a RAW format and unsuitable for visualisation. Typically, they go through an Image Signal Processing (ISP) pipeline, creating a display-adapted RGB non-linear image before being forwarded to the application-specific tool. In AVs, the ISP-processed RGB data can be used for tasks such as lane tracking [4], detection [5], segmentation [6], [7] and localisation [8]. Conventional hardware ISPs consist of separate functional blocks, each having a customised loss function independent of the others, leading to error propagation through the stages and significant error aggregation at the ISP output. This problem can be rectified by deep learning counterparts, though at the cost of increased complexity [9]–[11]. Additionally, the ISP introduces latency, higher power consumption (an ISP consumes 250 mW for 1080p video at 60 fps, while an object detection chip uses only 58.6 mW for $1080p$ at 30 fps), and an escalation in computational cost. These factors are extremely adverse for resource-constrained real-time systems, such as the AVs [12]–[15].

Traditionally, ISP pipelines have been designed and optimised for human visual perception. Research suggests that images of higher photographic quality, suitable for human perception, do not automatically ensure optimal CV performance, therefore, ISPs must be properly tuned to produce a suitable output for machine vision [18]–[20]. When a single ISP serves both human and computer vision systems, optimisation is usually geared towards meeting the quality standards of human vision, which may not guarantee optimal CV performance [18]. For this reason, there has been increasing interest in research aimed at refining ISP optimisation with a particular focus on CV as the final task [18], [21]–[23]. In scenarios where the same visual input is required for both human observation and machine perception, two distinct ISP pipelines may be necessary – one optimised for photographic quality and the other for real-time processing. Maintaining two separate ISPs introduces redundancy, increases latency, and complicates system integration. This motivates the exploration of ISP strategies that either unify processing or bypass it for CV applications

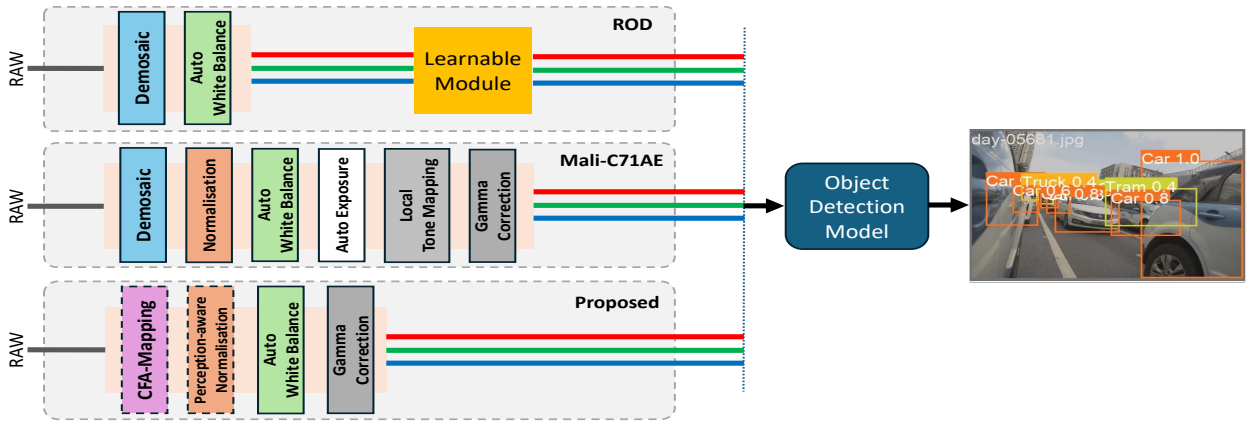


Fig. 1. A comparative overview of the three ISPs. ROD [16] (top), a baseline ISP inspired by ARM Mali-C71AE hardware (middle) [17] and the proposed simplified, computationally inexpensive ISP (bottom) is presented. All three ISPs involve statistical pre-processing. ROD utilizes an additional module with learnable parameters. The 3-channel RGB non-linear image is passed on to the object detection model.

altogether. This issue is particularly critical in safety-critical real-time applications, such as the perception systems in AVs, where the primary focus is on optimising the trade-off between accuracy and latency. In this work, we show how a straightforward and lightweight method can not only achieve a good balance between these factors but also ensure visually satisfactory outputs, despite not being specifically designed for human visualisation.

The key contributions of our work are as follows:

- Integrating perception-aware normalisation using the 99th percentile with Colour Filter Array (CFA) mapping, ensuring robust exposure handling across diverse lighting conditions while avoiding the over- or under-exposure issues commonly associated with standard normalisation techniques in deep learning models.
- A comparative study demonstrating that the proposed ISP achieves comparable or superior performance to state-of-the-art ISPs while maintaining a good trade-off between detection accuracy and preprocessing latency.

II. Related Work

To address the limitations of traditional ISP pipelines in CV applications, several studies have explored ISP models optimised for machine perception. Wu et al. [21] propose VisionISP, a trainable ISP pipeline tailored for machine vision, which optimises feature retention while reducing unnecessary processing overhead. Their approach emphasises content-aware processing to preserve details critical for object detection. Similarly, Shi et al. [20] explore an evolutionary algorithm-based optimisation framework for tuning ISP configurations specifically for high-level vision tasks. Their results show that an ISP optimised for human perception is not necessarily beneficial for CV and that an automated configuration search can significantly improve efficiency. Chen et al. [10] present a lightweight deep learning-based ISP model that replaces traditional ISP pipelines by employing a U-Net architecture with fine-grained attention and contextual upsampling. Although achieving a 23× parameter reduction and an 81× increase in computational efficiency, it remains better suited for mobile applications. Collectively, these studies highlight that designing ISPs specifically for vision tasks, rather than for human consumption, can lead to improved detection accuracy. However, they often rely on complex learning-based transformations, which are suboptimal for real-time perception systems.

To improve efficiency, some studies focus on hardware-centric lightweight ISP pipelines, while others explore software-based approaches. Datta et al. [24] develop an ISP-less vision system that processes RAW sensor data directly using in-sensor and in-pixel computing, reducing bandwidth and power demands. Lubana et al. [25] propose a minimalistic ISP pipeline for deep learning accelerators, integrating gamma compression and pixel binning at the sensor level to optimise efficiency. On the software side, Lin et al. [14] propose an automated neural architecture search framework to design lightweight ISP models that balance high image quality and low computational complexity. Their approach achieves over 90% reduction in floating-point operations compared to state-of-the-art models while maintaining strong performance. Xu et al. [16] introduce an end-to-end approach that integrates a High Dynamic Range (HDR) preprocessing module for object detection, emphasising the importance of retaining dynamic range information in RAW sensor data. Inspired by these computationally efficient methods, our research focuses on developing a lightweight software-based ISP pipeline optimised for real-time perception in safety-critical applications.

III. Methodology

In this section, we present the concrete steps of our minimalistic and lightweight pipeline which operates on RAW images and is tailored towards enhancing the performance of CV systems. We analyse the role of each block and present the experiment details. We compare our proposed method with two other state-of-the-art ISPs, the ROD model and a Baseline ISP. Since the ISP-processed images of the ROD dataset [16]

are not publicly available, we designed a Baseline ISP pipeline (see Figure 1). To ensure a realistic evaluation, we implemented a software emulation of the functional blocks of the ARM Mali-C71AE hardware ISP, a processor designed for real-time automotive applications [17].

The Baseline ISP pipeline is comprised of:

- Demosaicing
- Normalisation
- Auto White Balance (AWB)
- Exposure Adjustment
- Local Tone Mapping
- Gamma Correction

A. *Proposed Preprocessing Pipeline*

Figure 1 bottom shows the proposed ISP. To address the challenges posed by single-channel input and maintain real-time feasibility for automotive applications, we propose a computationally efficient preprocessing pipeline. Conventional demosaicing algorithms rely on interpolation to reconstruct missing colour information, therefore introducing computational overhead that may hinder real-time processing. Furthermore, interpolation leads to colour approximation to maintain spatial resolution resulting in introduction of noise in RAW data. Instead, we adopt a Bayer CFA rearrangement, where we map colour channel information using the RGGB CFA pattern. The two green channels are averaged to produce a 3-channel representation instead of a 4-channel one. This method provides a lightweight alternative that preserves essential colour structure without interpolation.

Once the 1-D RAW data is converted to a 3-D RGB linear image we normalise the data for further processing. We observe that the image data encompasses a diverse range of scenes with wide dynamic range variations. A standard normalisation technique involves using the maximum intensity of the signal. However, we draw inspiration from the lightness perception framework [26] to normalise a scene based on the perceivable 'white-point' instead of the maximum intensity to mitigate challenging cases of self-illuminating regions and establish a more data-driven normalised signal. Contrary to Krawczyk et al. [26] who adopted 95th percentile as the white point for tone mapping, we empirically set the 99th percentile (p_{99}) as the perceptual maximum, which helps suppress outliers while retaining meaningful contrast.

The normalised image I_{norm} is given by:

$$I_{\text{norm}}(x, y) = \min\left(\frac{I(x, y)}{p_{99}}, 1\right) \tag{1}$$

where:

- $I(x, y)$ is the linear image intensity at pixel (x, y).
- p_{99} is the 99th percentile value of the image.
- $I_{\text{norm}}(x, y)$ is the normalized image.
- The function $\min(\cdot, 1)$ ensures that values greater than 1 are clipped.

The Bayer pattern emphasises the green channel due to its higher sensitivity to human vision. Consequently, the simple

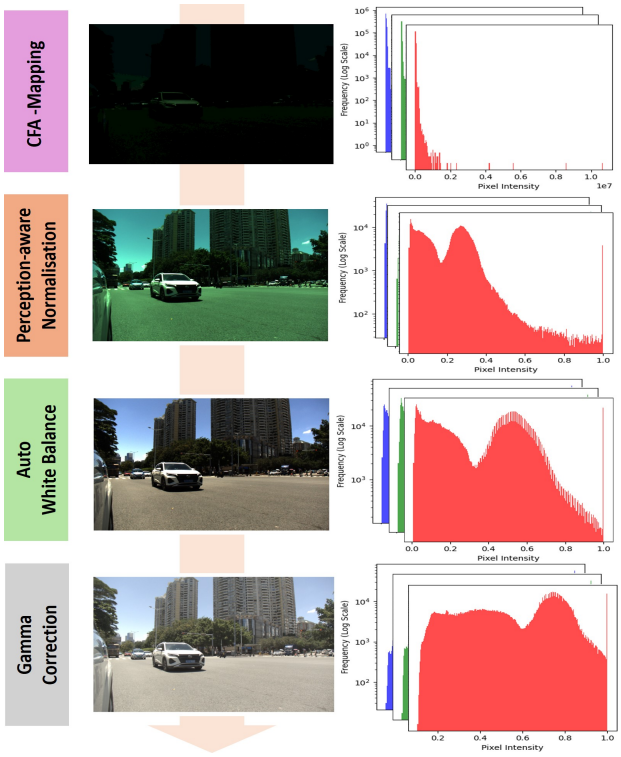


Fig. 2. The impact of each functional block on the frames is illustrated in terms of both visual appearance and intensity distribution. The transformation progressively adjusts the intensity distribution, evolving from a skewed distribution to a more uniform one in the non-linear space.

CFA-mapping has a green colour shift, which needs to be corrected. We apply a white balance correction to the normalised image I_{norm} as follows:

$$I'_{norm_c}(x, y) = \begin{cases} I_{norm_{\text{R}}}(x, y) \times \frac{\mu_G}{\mu_R}, & \text{if } c = R \\ I_{norm_{\text{G}}}(x, y), & \text{if } c = G \\ I_{norm_{\text{B}}}(x, y) \times \frac{\mu_G}{\mu_B}, & \text{if } c = B \end{cases} \tag{2}$$

where:

- $I_{norm_c}(x, y)$ and $I'_{norm_c}(x, y)$ are the original and white-balanced normalised intensities of channel c (where $c \in \{R, G, B\}$) at pixel (x, y).
- μ_R, μ_G, μ_B are the mean intensities of the red, green, and blue channels across the entire normalised image.

Post AWB correction the image is still linear. To match human visual perception and improve display adaptiveness a gamma-curve (2.2) correction is typically applied to transform the linear image to a perceptually uniform, non-linear space. Consequently, the gamma corrected image I_{g} enhances visibility by aligning the luminance distribution with human perception, ensuring balanced contrast in diverse lighting conditions, which improves object detection by preserving details in both dark and bright regions.

$$I_g(x, y) = I'_{norm}(x, y)^{\gamma} \tag{3}$$

where:

- $I'_{norm}(x, y)$ is the linear, normalised, white balance corrected image.
- $I_g(x, y)$ is the gamma-corrected image.
- γ is the gamma value, typically $(1/2.2)$

Through extensive ablation studies, we determined that the combination of CFA-mapping, AWB, p_{99} normalisation, and gamma correction provided the best balance between computational efficiency and detection performance. Through the introduction of each of the blocks we observe an incremental enhancement in image quality, which is validated by the experiment results (see Figure 2). We draw inspiration from Krawczyk et al. [26] to normalise the HDR images. Instead of p_{95}, we empirically use p_{99} to reduce clipping and better preserve the HDR information in the image. Interestingly, while conventional preprocessing steps, such as full demosaicing, normalisation by the maximum value, and local tone mapping, are widely used in standard CV pipelines, they offered minimal performance improvement over our optimised approach.

B. Dataset

One of the primary challenges in developing algorithms using RAW image data for automated driving applications is the scarcity of high-quality RAW image datasets. The ROD dataset [16] is a comprehensive dataset specifically designed to address this limitation by capturing both day and night driving scenarios with high dynamic range. Unlike other RAW datasets (PASCALRAW [27], LOD [28]) which lack automotive-grade sensor setups, diverse driving conditions, or sufficient HDR bit depth, ROD uniquely provides 24-bit images captured from vehicle-mounted cameras across day, night, and tunnel scenes, rendering it more suitable for this study. The dataset consists of 25,000 day and night RAW images (16,000 images have been publicly released), making it a valuable resource for training and evaluating object detection algorithms on HDR RAW sensor data.

C. Implementation Details

Our experiments were conducted on an NVIDIA Quadro RTX 5000 GPU with a batch size of 8. The dataset consisted of 4,053 day images and 12,026 night images. Each experiment was run for 300 epochs to evaluate the performance of the proposed preprocessing techniques. All preprocessing pipelines were implemented on the GPU, and the execution times refer exclusively to GPU processing time.

IV. Results

In this section, we evaluate the performance of three different preprocessing pipelines on day and night scenes from the ROD dataset. A custom split of 80%-10%-10% was made for the train-val-test cycles and we evaluate the performance over 609 day and 1805 night test images. For completeness, a fourth pipeline (Linear 3-channel) is also included in the evaluation. It directly passes the CFA-mapped three-channel image to the object detector without any preprocessing to ensure training on a linear RGB image. Table I presents the results from the experiments.

In our first method, 1-channel RAW images converted to 3-channel RGB images via CFA-Mapping, are passed on to the object detection model. The second method depicts our in-house implementation of the Baseline ISP pipeline where the functional blocks have been inspired by the real-time automotive ISP ARM Mali-C71AE [17]. The ROD pipeline [16] entails a minimal statistical preprocessing (demosaicing and AWB) and a learning-based module which implements local and global tone mapping. Finally, we compare our proposed pipeline which consists of simple computational blocks and minimalistic processing. The details of the ISPs are as presented in Figure 1 and Section III.

Since we target not only high safety but also real-time performance, Table I records the performance of the ISPs in terms of mean Average Precision (mAP) at IoU thresholds $[50 - 95]$ and 50, along with their respective preprocessing times. Furthermore, we make splits of the test dataset based on lighting conditions. *Day* and *Night* scenes provide different dynamic ranges and hence different challenges for object detection.

We observe that while reshuffling RAW sensor data into three channels via CFA-mapping requires minimal computational time (0.19 ms for day images and 0.2 ms for night images), it results in significantly poor detection accuracy under both day and night conditions. Without any preprocessing, this result follows our intuition as a linear image with a high dynamic range luminance distribution is unsuitable for human vision perception. Object detection modules such as YOLOv9 are adapted to image data suitable for human vision. Simple CFA-mapping is unable to enhance the RAW data enough to achieve a good performance.

Under daylight conditions, both the Baseline ISP and our Proposed ISP pipeline achieve approximately 10% higher mAP[50-95] scores and 6.5% higher mAP50 scores compared to the ROD ISP. The Baseline and Proposed methods perform similarly, with each slightly outperforming the other in different mAP cases. However, it must be reiterated that the Baseline ISP has significantly higher computational cost owing to the much elaborate statistical computational blocks (refer Figure 1). When comparing preprocessing times, we observe that ROD (2.6 ms) and our method (2.7 ms) have similar execution speeds due to their computationally lightweight pipelines. In contrast, the Baseline ISP introduces significant latency. However, our proposed method considerably outperforms ROD in detection accuracy.

Experiments under night-time lighting conditions reveal some interesting observations. The Baseline and Proposed ISPs exhibit very similar detection performance for both mAP50 and mAP[50-95], but the Baseline is more than 2 times slower. ROD ISP narrowly outperforms the proposed method in preprocessing time by 0.2 ms while achieving a superior mAP50 score. Although our proposed method lags behind ROD by 3.2% in mAP50, it surpasses it by 5.5% in mAP[50-

TABLE I
QUANTITATIVE COMPARISON BETWEEN ISPS IN TERMS OF DETECTION ACCURACY AND EXECUTION TIME. THE BEST ACCURACY RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED.

Lighting Condition	Method	mAP[50-95]	mAP50	Preprocessing Time (ms)
Day	Linear (3-channel)	22.4%	33.0%	0.19
	Baseline ISP	<u>40.8%</u>	**57.8%**	8.00
	ROD	31.8%	51.0%	2.60
	Proposed	**41.0%**	<u>57.5%</u>	2.70
Night	Linear (3-channel)	15.5%	25.9%	0.20
	Baseline ISP	**55.6%**	<u>77.5%</u>	7.00
	ROD	50.5%	**80.7%**	2.50
	Proposed	<u>55.0%</u>	<u>77.5%</u>	2.70

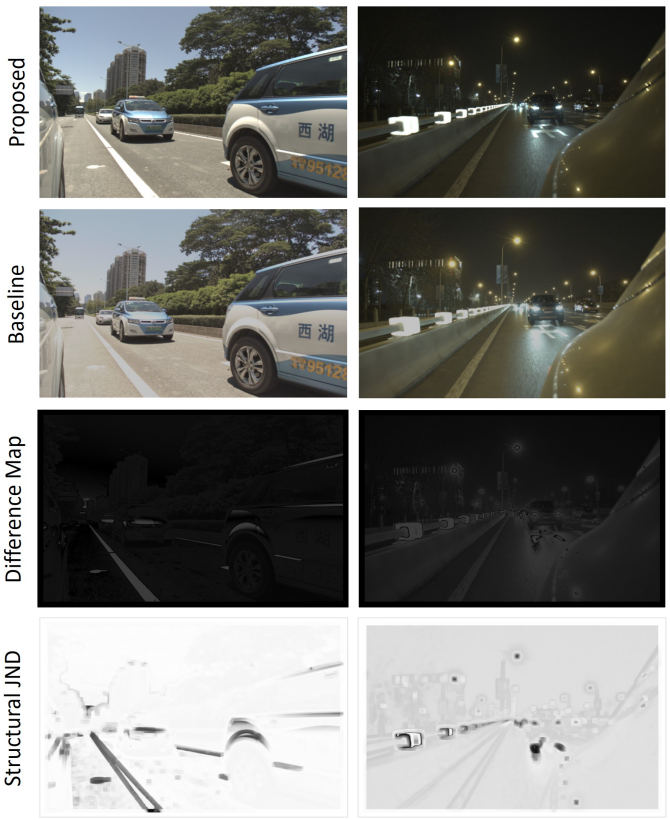


Fig. 3. Intensity difference map and *Just Noticeable Difference* (JND) [29] map based on structural similarity between images created by proposed ISP and Baseline ISP. Images produced by the two ISPs from day and night lighting conditions show visible closeness. Images from both ISPs achieve better object detection accuracy. However as mentioned in Table I, proposed method achieves the same at a significant lower cost.

95] while maintaining similar computational efficiency. The superior performance of ROD can be attributed to its additional learnable module, which is capable of locally enhancing night scenes using a data-driven approach.

V. DISCUSSION

Weighing all the facets of comparison, our proposed ISP achieves an effective balance between detection accuracy and real-time performance. Unlike the baseline ISP, which relies on a full stack of traditional processing blocks, our method achieves similar mAP scores with a fraction of the computational cost, making it better suited for deployment in latency-critical automotive systems. Figure 3 shows how the output of our Proposed ISP is visibly very close to the Baseline. We compute intensity difference maps and MSSIM-based [29] *Just Noticeable Difference* (JND) maps to show overall similarity in high-contrast regions of the image. Unlike learning-based tone mapping, which is typically optimised over the training dataset, our method adapts dynamically to each input frame using perception-aware normalisation. In cases of extreme lighting conditions, using self-luminous pixels with maximum intensity to normalise the image can lead to improper compression [26]. However, p_{99} provides a more stable and task-relevant white point that avoids clipping and preserves contrast and structural information, critical for downstream object detection. This ensures that exposure and contrast adjustments are optimised for each frame, avoiding issues such as over-saturation in bright scenes or excessive darkening in low-light environments.

Limitations: While this study demonstrates the benefits of a computationally efficient ISP for object detection, several limitations must be acknowledged. First, the absence of publicly available RGB outputs from the ROD dataset and the incompatibility of most open-source ISPs with 24-bit HDR RAW inputs required the development of a custom baseline ISP. Although carefully designed to emulate real-world pipelines, this baseline may not fully capture the complexity of industrial-grade ISPs. Second, the limited availability of diverse automotive RAW datasets restricts the generalisability of our findings, as evaluation on a single dataset cannot fully account for variations across different sensor types and driving conditions. Third, despite achieving low overall latency, the computational burden introduced by perception-aware normalisation remains significant and could hinder real-time deployment in resource-constrained environments.

VI. CONCLUSION

In this work, we have demonstrated that a carefully designed combination of lightweight preprocessing techniques

can effectively balance computational efficiency and object detection performance for AV applications. By integrating CFA-based mapping with perception-aware normalisation, we eliminate full demosaicing while still ensuring robust exposure handling across diverse lighting conditions. Normalising to the 99th percentile proves particularly beneficial, as it mitigates the influence of extreme intensity values, ensuring a more stable and adaptive response to challenging lighting conditions commonly encountered in automated driving. Our experimental results show that the proposed approach achieves comparable or superior detection accuracy to conventional ISP pipelines while significantly reducing computational overhead, rendering it highly suitable for real-time, resource-constrained automotive perception systems.

Future work could focus on optimising the estimation of p_{99} to reduce its computational cost, for example through histogram-based approximations or adaptive sampling. Beyond latency improvements, expanding the evaluation to include additional RAW datasets and peer-reviewed ISP architectures would extend the generalisability of the proposed approach. Incorporating lightweight learnable modules into the pipeline may also allow for task-adaptive tuning of preprocessing operations while preserving interpretability and efficiency. Finally, evaluating the integration of the proposed pipeline with multiple object detectors and assessing its performance on compute-constrained platforms would further demonstrate its practical value for deployment in real-time automotive applications.

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